

2.

Izračunaj komutatorje Paulijevih matrik

$$[A, B] = AB - BA \quad \sigma_x = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \sigma_y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \quad \sigma_z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$[\sigma_x, \sigma_y] = \begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix} - \begin{bmatrix} -i & 0 \\ 0 & i \end{bmatrix} = \underline{2i \sigma_z}$$

$$[\sigma_y, \sigma_z] = \begin{bmatrix} 0 & i \\ i & 0 \end{bmatrix} - \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} = \underline{2i \sigma_x}$$

$$[\sigma_z, \sigma_x] = \underline{2i \sigma_y}$$

3. Poišči vse matrike $x \in \mathbb{R}^{2 \times 2}$, ki rešijo enačbo

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \cdot x \cdot \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 2 \\ 1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a+c & b+d \\ c & d \end{bmatrix}$$

$$\begin{bmatrix} a+c & b+d \\ c & d \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} a+b+c+d & b+d \\ c+d & d \end{bmatrix} = \begin{bmatrix} 3 & 2 \\ 1 & 0 \end{bmatrix}$$

$$\begin{aligned} d &= 0 \\ c+d &= c = 1 \\ b+d &= b = 2 \\ a+b+c+d &= a+3 = 3 \\ a &= 0 \end{aligned}$$

$$\underline{\underline{x = \begin{bmatrix} 0 & 2 \\ 1 & 0 \end{bmatrix}}}$$

SLGD:

$$\text{tr} A = A_{11} + A_{22} + \dots + A_{nn} = \sum_{k=1}^n A_{kk}$$

$$\text{tr}(A+B) = \text{tr} A + \text{tr} B$$

$$\text{tr}(cA) = c \text{tr} A$$

$$\text{tr}(A \cdot B \cdot C) = \text{tr}(B \cdot C \cdot A) = \text{tr}(C \cdot A \cdot B)$$

④ A, C obrnjeni kvad. mat.; poišči inverz matrike

$$D = \begin{bmatrix} A & B \\ 0 & C \end{bmatrix}$$

$$D \cdot D^{-1} = I$$

$$\begin{bmatrix} A & B \\ 0 & C \end{bmatrix} \begin{bmatrix} a' & b' \\ c' & d' \end{bmatrix} = \begin{bmatrix} I & 0 \\ 0 & I \end{bmatrix}$$

$$Aa' + Bc' = I \Rightarrow \underline{a' = A^{-1}}$$

$$Ab' + Bd' = 0 \Rightarrow Ab' + BC^{-1} = 0 \Rightarrow b' = -A^{-1}BC^{-1}$$

$$Cc' = 0 \Rightarrow \underline{c' = 0}$$

$$Cd' = I \Rightarrow \underline{d' = C^{-1}}$$

$$\underline{\underline{D^{-1} = \begin{bmatrix} A^{-1} & -A^{-1}BC^{-1} \\ 0 & C^{-1} \end{bmatrix}}}$$

RKF

- računanje ranga

Gaussova eliminacija

$$- v_i \leftarrow \alpha v_i$$

$$- v_i \leftarrow v_i + \beta v_j$$

$$- v_i \leftrightarrow v_j$$

⑥ Prevedi matriki v VKF in izračunaj rang

a)

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & -1 & 0 \\ 2 & 1 & 0 \\ -1 & 1 & 1 \\ 1 & 3 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 \\ 0 & -1 & 0 \\ 0 & -1 & -2 \\ 0 & 2 & 2 \\ 0 & 2 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$rk(A) = 3$$

$$b) \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & -1 \\ 1 & 0 & 2 & 1 \\ 0 & 0 & -1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \text{rank}(A) = 3$$

⑤ Inverz

$$\left[A \mid I \right] \rightarrow \left[I \mid B \right]$$

$\downarrow$
 $= A^{-1}$

$$a) A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 3 \\ 1 & 3 & 1 \end{bmatrix}$$

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 2 & 1 & 3 & 0 & 1 & 0 \\ 1 & 3 & 1 & 0 & 0 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & -3 & 1 & -2 & 1 & 0 \\ 0 & 1 & 0 & -1 & 0 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 0 & 1 \\ 0 & 0 & 1 & -5 & 1 & 3 \end{array} \right] \rightarrow$$

$$\rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 8 & -1 & -5 \\ 0 & 1 & 0 & -1 & 0 & 1 \\ 0 & 0 & 1 & -5 & 1 & 3 \end{array} \right]$$

$$b) A = \begin{bmatrix} 1 & x & z \\ 0 & 1 & y \\ 0 & 0 & 1 \end{bmatrix} \quad x, y, z \in \mathbb{R}$$

$$\left[\begin{array}{ccc|ccc} 1 & x & z & 1 & 0 & 0 \\ 0 & 1 & y & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & z-xy & 1 & -x & 0 \\ 0 & 1 & y & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & -x & xy-z \\ 0 & 1 & y & 0 & 1 & -y \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right]$$

Heisenbergova algebra: $\begin{bmatrix} 1 & x & z \\ 0 & 1 & y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & x' & z' \\ 0 & 1 & y' \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & x+x' & z+z'+xy' \\ 0 & 1 & y+y' \\ 0 & 0 & 1 \end{bmatrix}$

na $H = \{(x, y, z); x, y, z \in \mathbb{R}\}$ definirano $(x, y, z) +_H (x', y', z') = (x+x', y+y', z+z'+xy')$

- enota $(0, 0, 0)$

- ni komutativna

- naspr. el. $(-x, -y, -z+xy)$

- Heisenbergova grupa

